

Assessing Infrastructure Vulnerability to Natural Hazards in South Tyrol

Roland Bernard, 19598, rolbernard@unibz.it

February 2026

1 Domain of Interest

The idea of the project is to focus on the intersection of exposed assets and natural hazards. The aim is to identify and analyze critical infrastructure (water, electricity, and communications) that are geographically located within known hazard zones (landslides and avalanches) to assess potential vulnerabilities.

2 Data Sources

The project will integrate five datasets from the Bolzano Province GeoKatalog. The integration will involve the following provided data sources:

- **Infrastructure:** bz_infrastructure_line and bz_infrastructure_node will be utilized to provide information regarding critical infrastructure.
- **Hazards:** Information regarding known risk areas will be derived from bz_hazard_landslide and bz_hazard_avalanche.
- **Administrative:** bz_admin_municipality to provide spatial context and assign risk to specific municipalities.

3 Application and Queries

By integrating infrastructure assets with hazard inventories and administrative data, the system will be able to answer the following types of queries:

- Which water-related infrastructure nodes are located within high risk landslide zones?
- What are the infrastructure lines that intersect with avalanche-prone areas?
- In a given Municipality, how many electricity lines are currently situated in a hazard zone.
- Which municipalities have the highest count of infrastructure elements exposed to natural hazards?

4 Initial Ontology

An OWL 2 QL ontology will be created that contains relevant classes (e.g., Infrastructure, EnergyInfrastructure, HazardZone, Municipality) and properties (e.g., isExposedTo, infrastructureIn). A preliminary design for the ontology is shown in Fig. 1.

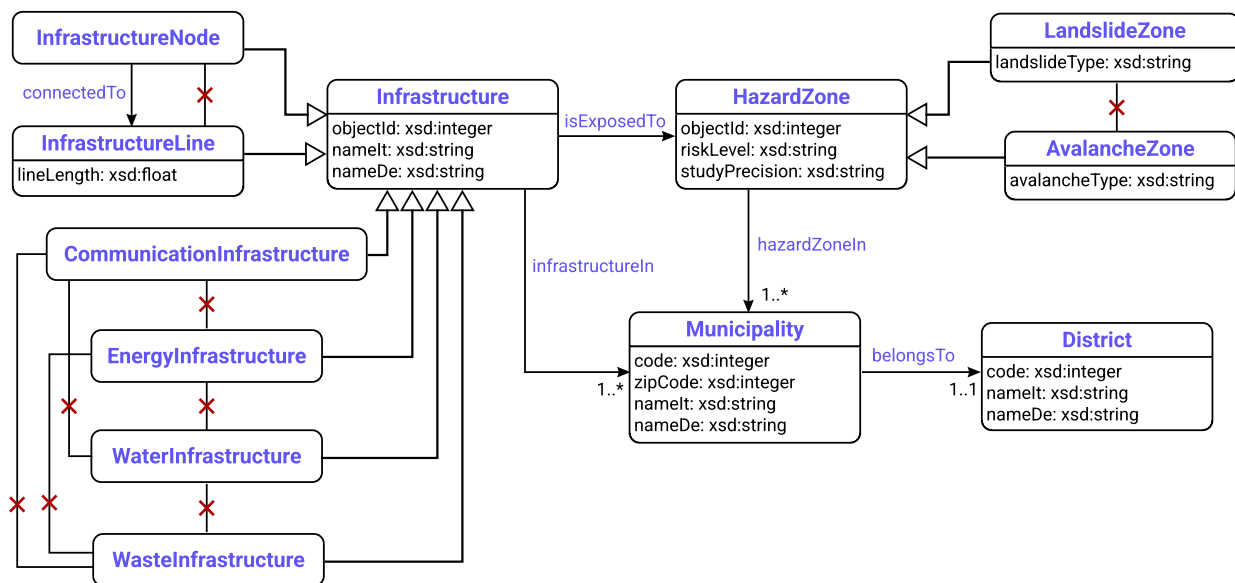


Figure 1: Proposed ontology for the project.